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IMPLEMENTATION COMPLETION AND RESULTS REPORT

LOAN 8484-AR

ON A LOAN

IN THE AMOUNT OF US\$ 200 MILLION

TO THE

Argentine Republic

FOR A

Renewable Energy for Rural Areas Project
June 28, 2024

Energy & Extractives Global Practice
Latin America And Caribbean Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective {June 11, 2024})

Arg.\$ 901 = US\$1

FISCAL YEAR
July 1 - June 30

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ABBREVIATIONS AND ACRONYMS

BNA	Central Bank of Argentina (<i>Banco de la Nación Argentina</i>)
CERC	Contingency Emergency Response Component
CPF	Country Partnership Framework
CPS	Country Partnership Strategy
DiGePPSE	General Directorate of Sectoral and Special Programs and Projects (<i>Dirección General de Programas y Proyectos Sectoriales y Especiales</i>)
EIRR	Economic internal rate of return
ESMF	Environmental and Social Management Framework
FM	Financial Management
GoA	Government of Argentina
GRM	Grievance Redress Mechanism
IBRD	International Bank for Reconstruction and Development
ICR	Implementation Completion and Results Report
INTA	National Institute for Agriculture and Livestock Technology (<i>Instituto Nacional de Tecnología Agropecuaria</i>)
IPPF	Indigenous Peoples Planning Framework
ISRs	Implementation Status and Results Reports
M&E	Monitoring and Evaluation
MW	Mega-Watt
MTR	Mid-Term Review
NPV	Net present value
O&M&R	Operation&Maintenance&Replacement
OM	Operational Manual
PAD	Project Appraisal Document
PCU	Project Coordinating Unit
PDO	Project Development Objective
PERMER I	Argentina Renewable Energy Markets Project (<i>Proyecto de Energías Renovables en Mercados Rurales I</i>), P006043 and P045048 (grant)
PERMER II	Argentina Renewable Energy Markets Project (<i>Proyecto de Energías Renovables en Mercados Rurales II</i>), P133288
PEU	Provincial Executing Unit
PV	Photovoltaic
RPF	Resettlement Policy Framework
SE	Secretariat of Energy (<i>Secretaría de Energía</i>)
SHS	Solar home system
SSEE	Deputy Secretariat for Electricity (<i>Subsecretaría de Energía Eléctrica</i>)
WTP	Willingness-to-pay

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DATA SHEET

BASIC INFORMATION

Product Information

Project ID	Project Name
P133288	Argentina Renewable Energy for Rural Areas Project
Country	Financing Instrument
Argentina	Investment Project Financing
Original EA Category	Revised EA Category
Partial Assessment (B)	Partial Assessment (B)

Organizations

Borrower	Implementing Agency
Argentine Republic	Secretaria de Energía

Project Development Objective (PDO)

Original PDO

The project's development objective is to provide and enhance access to modern energy services in selected rural areas of Argentina.

**FINANCING**

	Original Amount (US\$)	Revised Amount (US\$)	Actual Disbursed (US\$)
World Bank Financing			
IBRD-84840	200,000,000	91,511,634	91,511,634
Total	200,000,000	91,511,634	91,511,634
Non-World Bank Financing			
Borrower/Recipient	10,850,000	0	0
Local Govts. (Prov., District, City) of Borrowing Country	5,740,000	0	0
Local Sources of Borrowing Country	23,500,000	0	0
Total	40,090,000	0	0
Total Project Cost	240,090,000	91,511,634	91,511,634

KEY DATES

Approval	Effectiveness	MTR Review	Original Closing	Actual Closing
07-Apr-2015	04-May-2016	09-May-2019	31-Oct-2020	31-Dec-2023



RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Amount Disbursed (US\$M)	Key Revisions
29-Oct-2020	44.86	Change in Results Framework Change in Components and Cost Change in Loan Closing Date(s) Cancellation of Financing Reallocation between Disbursement Categories Change in Implementation Schedule Other Change(s)
23-Jun-2022	69.26	Change in Results Framework Change in Components and Cost Change in Loan Closing Date(s) Reallocation between Disbursement Categories Change in Implementation Schedule
11-Jun-2023	91.01	Change in Results Framework Change in Components and Cost Change in Loan Closing Date(s) Cancellation of Financing Reallocation between Disbursement Categories Change in Implementation Schedule

KEY RATINGS

Outcome	Bank Performance	M&E Quality
Moderately Unsatisfactory	Satisfactory	Substantial

RATINGS OF PROJECT PERFORMANCE IN ISRs

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US\$M)
01	30-Dec-2015	Moderately Satisfactory	Moderately Satisfactory	0
02	30-Jun-2016	Moderately Satisfactory	Moderately Satisfactory	20.00
03	11-Jan-2017	Moderately Satisfactory	Moderately Satisfactory	20.00
04	29-Aug-2017	Moderately Satisfactory	Moderately Unsatisfactory	20.00
05	16-Jan-2018	Moderately Satisfactory	Moderately Unsatisfactory	20.00
06	20-Sep-2018	Moderately Satisfactory	Moderately Unsatisfactory	20.00



07	18-Apr-2019	Moderately Unsatisfactory	Moderately Unsatisfactory	26.10
08	05-Nov-2019	Moderately Unsatisfactory	Moderately Satisfactory	38.01
09	18-May-2020	Moderately Unsatisfactory	Moderately Satisfactory	44.86
10	27-Oct-2020	Moderately Satisfactory	Moderately Satisfactory	44.86
11	29-Apr-2021	Moderately Unsatisfactory	Moderately Unsatisfactory	51.86
12	19-Nov-2021	Moderately Unsatisfactory	Moderately Unsatisfactory	62.36
13	30-Jun-2022	Moderately Satisfactory	Moderately Satisfactory	73.26
14	21-Dec-2022	Moderately Satisfactory	Moderately Satisfactory	91.01
15	21-Jul-2023	Moderately Satisfactory	Moderately Satisfactory	91.01
16	19-Dec-2023	Moderately Unsatisfactory	Moderately Unsatisfactory	91.01

SECTORS AND THEMES

Sectors

Major Sector/Sector (%)

Energy and Extractives 100

Renewable Energy Solar 100

Themes

Major Theme/ Theme (Level 2)/ Theme (Level 3) (%)

Social Development and Protection 4

Social Inclusion 4

Indigenous People and Ethnic Minorities 2

Other Excluded Groups 1

Participation and Civic Engagement 1

Human Development and Gender 2

Gender 2



Urban and Rural Development	93
Rural Development	93
Rural Infrastructure and service delivery	93
Environment and Natural Resource Management	100
Climate change	100
Mitigation	100

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I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

Context

1. **At appraisal, Argentina had been one of the top performers in the region in reducing poverty and expanding the middle class, and the Government of Argentina (GoA) remained committed to promoting growth with equity and inclusion.** Total poverty had declined from 31 percent in 2004 to 10.8 percent in 2013, while extreme poverty had fallen from 17 to 4.7 percent. Income inequality had shown significant progress also, and the middle class grew by 68 percent between 2004 and 2012. At the same time, the GoA had set the goal of continued economic growth with social inclusion and improved competitiveness. This challenge required sustained growth and a focused and efficient deployment of public services geared to protect the most vulnerable.

2. **However, the strong economic growth of the previous decade had been accompanied by rising macro imbalances.** Challenges included inflationary pressures, deficits in fiscal and current accounts and limited international reserves. In spite of relatively modest fiscal and current deficits, and a low public sector debt, limited access to capital markets created pressures on the economy. These imbalances had to be resolved to ensure the sustainability of the gains in equity and development achieved during the previous decade.

3. **Private and public electricity generators were able to satisfy demand mostly through thermal generation, which in part relied on subsidized imported fuels.** Argentina's fully unbundled electricity system relied on thermal (60 percent of installed capacity) and hydropower (36 percent) facilities that satisfied the demand of 15 million customers (13 million residential and 1.5 million commercial). Reliance on subsidized imported fuels challenged the fiscal and current accounts balances of the country and created pressures on the Central Bank's reserves. In isolated and rural areas, the sector's reliance on fossil fuels was relatively minor. Subsidies for rural electrification were negligible and relied mostly on existing transfer schemes.

4. **Universal access to modern energy service was one of the GoA's main priorities.** While Argentina's level of electrification was high, a significant part of the population remained without access—or partial, inadequate access—to modern energy services, severely limiting their resilience to extreme poverty. The number of households without access was around 145,000, mostly living in rural areas, including an important number of indigenous communities.

5. **Under the leadership of the Secretariat of Energy (*Secretaría de Energía [SE]*), the GoA had implemented several projects aimed at increasing access to energy services in rural areas. Among them the Bank-financed PERMER I¹ project was the most successful.** This first project provided rural areas with sustainable and reliable electricity using mostly renewable energy technologies and supporting thermal applications and water pumping in public buildings. It addressed the needs of more than thirty thousand households in fifteen provinces proving to be a valuable mechanism to increase electricity access in rural areas. As an evolution to previous efforts, the proposed project (PERMER II) was designed to increase the coordination between the Federal Government and provinces, thus responding to the need of a common approach toward rural electrification.

¹ Argentina Renewable Energy Markets Project, PERMER I for its Spanish title: *Proyecto de Energías Renovables en Mercados Rurales I*.



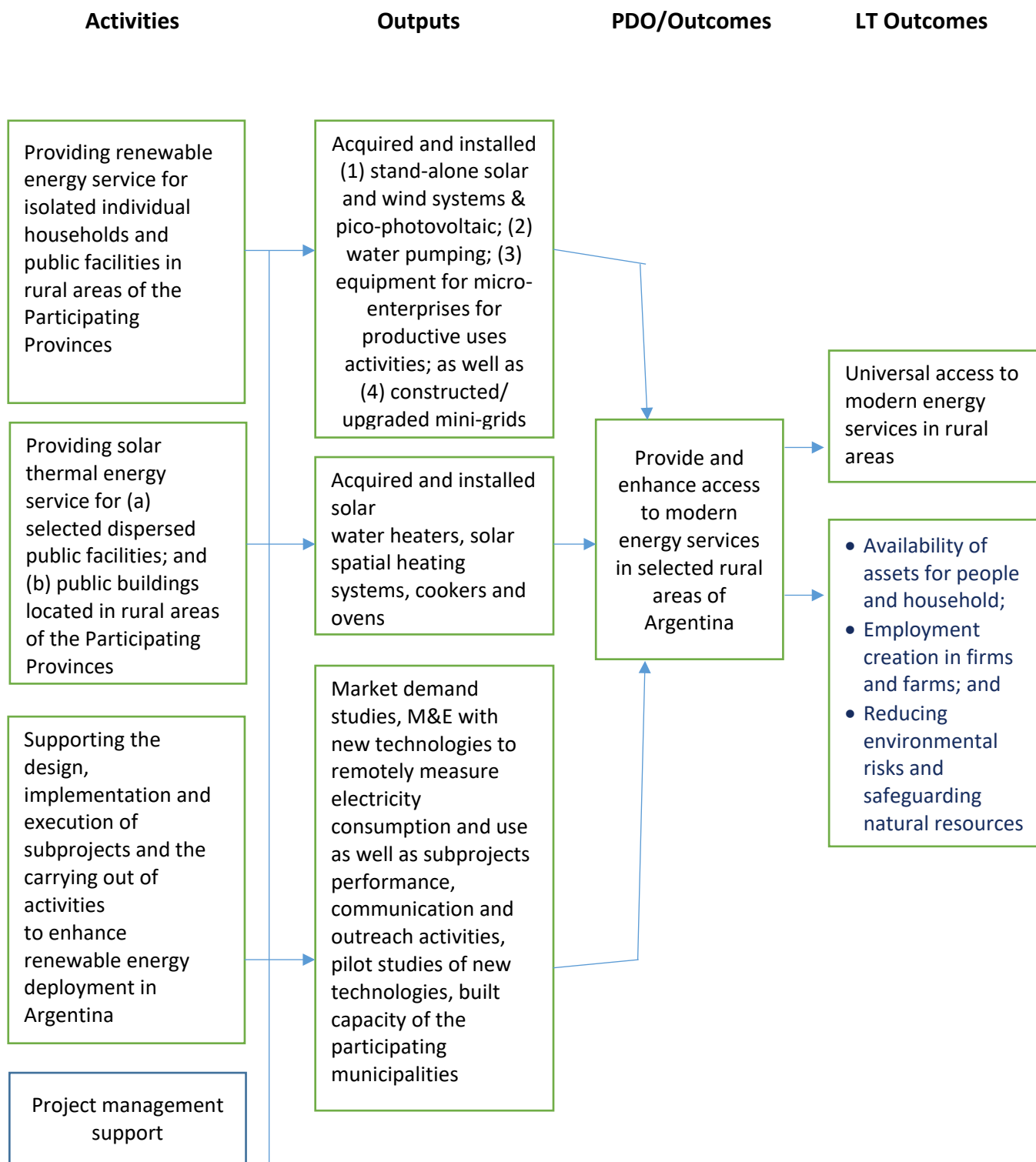
6. **The PERMER II was proposed as the main national instrument to move Argentina toward universal energy access.** In this respect, it was expected to provide or enhance the availability of energy infrastructure for rural households, helping the isolated rural population to leave extreme poverty behind. Also, the operation was aimed at supporting the Government's objectives to improve energy security, diversify the energy matrix and reduce environmental impacts.

7. **The project was fully aligned with the World Bank Group's Country Partnership Strategy (CPS 2015-18) for Argentina,** which focused on promoting shared prosperity and reducing poverty around three themes (a) employment creation in firms and farms; (b) availability of assets for people and households; and (c) reducing environmental risks and safeguarding natural resources. The project was designed to support directly the second theme by expanding asset availability for households in an efficient and sustainable manner, while also contributing toward the other two themes as economic uses of electricity enhance rural employment and the use of renewable energy helps reducing environmental risks.

8. **By the time of project appraisal, the World Bank had been playing an important role (since the late 1990s) in supporting the country's universal energy access effort.** The positive experience gained by both parties during this long partnership, justified the Bank's continued support on off-grid solutions, which was complementary to the activities financed by other international financial institutions.

Theory of Change (Results Chain)

9. **The project development objective (PDO) is expected to contribute to the fulfillment of the CPS for Argentina. Project components and activities were designed to support the achievement of the energy sector's rural electrification goals which are essential for reaching the PDO.** While the Project Appraisal Document (PAD) did not articulate an explicit theory of change or results chain, the theory of change corresponding to the formulated PDO and components is illustrated in the chart below. Project activities provide infrastructure and technical assistance to support directly an increase in the number of beneficiaries, obtaining new or enhanced access to energy services in rural areas, thus contributing to the PDO and the country's long term universal access objective. The achievement of the project's outcomes considers the following underlying assumptions and possible external factors: (i) that sustained political commitment in reducing rural poverty and foster inclusion will support the project's implementation and sustainability; (ii) possible macroeconomic imbalances will not have a significant impact on the project's execution nor on its achievements; and (iii) that the measures to address institutional weaknesses, including assistance to stakeholders and capacity building to strengthen the Project Coordinating Unit (PCU) and others entities, will succeed in dealing with these issues.





Project Development Objectives (PDOs)

10. The project's development objective is to provide and enhance access to modern energy services in selected rural areas of Argentina.

Key Expected Outcomes and Outcome Indicators²

- a) Direct project beneficiaries
- b) People provided with access to electricity by household connections
- c) Community electricity connections constructed under the project
- d) People that obtain daily and constant electricity service through mini-grids.

Components

11. The project comprised the following four components in rural areas of the participating provinces:

12. **Component 1: Renewable electricity service provision (US\$215.61 million, of which US\$183.40 million IBRD).** Provision of renewable electricity services through, among others (a) acquisition and installation of stand-alone solar and wind systems and pico-photovoltaic (PV) for isolated individual households and public facilities; (b) construction and/or upgrade of mini-grids for, among others, rural and isolated populations clusters and public institutions; (c) acquisition and installation of water pumping systems in isolated individual households, public facilities and isolated communities; (d) equipment to micro-enterprises for the development of productive uses activities; and (e) small works and goods as in-kind compensation under the relevant resettlement plans.

13. **Component 2: Solar thermal energy service provision (US\$9.67 million, of which US\$7.90 million IBRD).** Provision of solar thermal energy through the acquisition and installation of solar water heaters, solar spatial heating systems, cookers and ovens for selected disperse public facilities and public buildings.

14. **Component 3: Project deployment support (US\$6.71 million, of which US\$5.70 million IBRD).** To support the design and execution of sub-projects, and carrying out activities to enhance renewable energy deployment in the country, including: (a) markets studies on the potential energy demand and technologies; (b) M&E activities, including remotely measuring electricity consumption and sub-projects performance; (c) communication strategies and outreach activities; and (d) support of pilot studies for technologies not fully deployed in Argentina (e.g. solar spatial heating).

15. **Component 4: Project management (US\$8.10 million, of which US\$3.00 million IBRD).** Support to project management activities, including strengthening of the operating capacity of the PCU and Provincial Executing Units (PEUs), projects audits, training, travel costs and rental of vehicles for project activities.

16. **As shown in Annex 3, the actual allocation of resources differed from the original proposal, both for each component as well as for the total project cost.** As an average, 72.7 percent of the loan proceeds were actually disbursed and the ratio of disbursements per component ranged from 76.3 percent to zero, as component 2 was cancelled. In addition to IBRD financing, there was a local contribution of US\$4.8 million from the provincial government of Salta in financing Component 1.

² The four key outcome indicators were complemented by nine additional breakdown or supplemental indicators. See Annex 1.



B. SIGNIFICANT CHANGES DURING IMPLEMENTATION (IF APPLICABLE)

17. The project was subject to three restructurings, all of which involved changes in the targets of the Results Framework. Project restructurings encompassed also changes in scope, implementation period and partial cancellations, thus affecting expected outcomes and outputs.

Revised PDOs and Outcome Targets

18. No changes were made to the original PDO, but the outcome targets were revised, as shown in Table 1.

Table 1. Results Framework: PDO Indicators – Changes on End Targets

PDO Indicator	Original (PAD) end targets	Restructurings end targets		
		1 st (Oct. 2020)	2 nd (Jun. 2022)	3 rd (Jul. 2023)
Direct project beneficiaries	756,000	454,000	509,468	229,000
People provided with access to electricity by household connections	175,000	193,991	348,091	139,600
Community electricity connections constructed under the project	1,800	2,615	1,660	780
People that obtain daily and constant electricity service through mini-grids.	37,000	3,600	3,872	340

19. Overall, end targets were decreased considerably due to the partial cancellations of the loan (addressed below) and reduced expectations associated to significant delays caused by organizational changes in the government, the impact of the pandemic and, during the last two years, the negative impact of import restrictions that strongly affected the implementation of several subprojects.

20. It is noted that while the trend was to reduce the end targets of PDO indicators, some targets increased temporarily, the most noticeable being the number of people (households) provided with access to electricity, which increased by 79 percent during the second restructuring. Beside the factors causing target reductions (cancellations and delays), fluctuations in the end targets are explained by the government's shifting preferences on technology options to address electricity access, changing its focus from technologies that reach less household beneficiaries (but more community connections and a better quality of service, such as mini-grids) to those reaching a large number of beneficiaries per unit cost (small individual solar home systems – SHS), and vice versa.

Revised PDO Indicators

21. The PDO indicator definitions were not changed.

Revised Components

22. Component 2: Solar thermal energy service provision, was dropped, upon the GoA's proposal, during the second restructuring.



Other Changes

23. **Partial Cancellations.** The loan suffered two partial cancellations. An amount of US\$31.3 million was cancelled during the first restructuring. A second partial cancellation of US\$23.3 million was approved in the third restructuring.

24. **Component costs.** Table 2 presents the original project costs by components and how they changed, by restructuring.

Table 2. Project Costs by Components – Restructuring Changes (US\$ million)

	Original Allocation (PAD)	Restructurings		
		1 st (Oct. 2020)	2 nd (Jun. 2022)	3 rd (Jul. 2023)
Component 1: Renewable electricity service provision	183.40	160.539	164.121	140.841
Component 2: Solar thermal energy service provision	7.90	5.005	0	0
Component 3: Project deployment support	5.70	0.772	1.743	1.743
Component 4: Project management	3.00	1.843	2.295	2.295
Front-end fee	---	0.500	0.500	0.500
Total	200.00	168.659	168.659	145.379

25. **Reallocation between disbursement categories.** Restructurings implied a reallocation between categories. The most significant was the reallocation of the resources originally assigned to Component 2, which was cancelled during the second restructuring of mid-2022. Final disbursements per categories are presented in Annex 3.

26. **Closing date.** The Loan's closing date was extended by twenty months, twelve months and finally by six months, during the three project restructurings, for an accumulated extension of thirty-eight months. That is, closing date was extended from October 31, 2020 to December 31, 2023.

Rationale for Changes and Their Implication on the Original Theory of Change

27. **The project was restructured in three different situations.** The rationale for each case was as follows:

- Acknowledging the project's slow start, the first restructuring was justified by the need to enable the project to reach the estimated 450,000 beneficiaries, increase resilience of vulnerable and remote communities to the COVID-19 pandemic and improve local penetration of the project in provinces that had not been reached in that moment. The partial cancellation of the loan responded to the GoA's decision of redirecting IBRD and government resources to other projects aimed at the containment of the Covid-19 health pandemic. The need to extend the loan's closing date had been discussed already during the Mid-Term Review (MTR) in May 2019, as well as updating the targets of most indicators. In this respect, the first restructuring (effective in October 2020) came late. The causes of this significant delay were: (i) the year 2019 was characterized by the uncertainties of an election year that implied changes in the administration, including sector authorities, and the inherent difficulty in agreement on a restructuring that encompassed a partial cancellation of the loan; and (ii) the COVID-19 pandemic slowed considerably most project activities, including the decision-making process, during 2020. Annex 4 displays a project timeline that identifies multiple events that affected the implementation of the project.



- (b) The second restructuring was justified essentially by the need to extend the project's closing date. Given the considerable delays caused by numerous changes in the sector's organization and authorities, and the lockdown associated to the pandemic, it was deemed necessary to extend the loan closing date in order to address effectively a high priority electrification project. Given the time and human resources constraints the Secretary of Energy decided to focus its efforts on the electrification component and supporting activities. Therefore, it was decided not to allocate any resources to Component 2 and that the loan proceeds originally allocated to it were to be made available for the other components.
- (c) The third restructuring was needed to extend the project closing date to continue the implementation of critical ongoing sub-projects and achieve the PDO. Import restrictions had affected severely many project activities and fiscal constraints prevented the government to have the resources needed to cover the gap in case the loan was not extended. The second partial cancellation was approved as it was understood that such amount would not be committed and implemented by project closing.

28. The changes made did not have an impact on the project's original theory of change with exception of one specific activity, Component 2, that was dropped, but this had a minor impact on the achievement of outputs as it was a smaller component and the PDO indicators' targets, were revised to reflect more accurately the new conditions.

II. OUTCOME

A. RELEVANCE OF PDOs

Assessment of Relevance of PDOs and Rating

29. **The relevance of the PDO was maintained through the entire implementation period as top priorities of the Country Partnership Framework (CPF) for 2019-22 involved closing the gap in the provision of basic infrastructure services and transitioning to a cleaner energy matrix.** Also, the project continued matching the World Bank Group's selectivity criteria which included the country's demand and the Bank's comparative advantage, i.e. a strong track record in rural electrification. Project relevance was confirmed by the 2022 Performance and Learning Review of the CPF which established the objective of improving access to basic service delivery in vulnerable areas, an objective directly linked to the rural electrification effort. **Therefore, the project's relevance is rated High through the whole implementation process.**

B. ACHIEVEMENT OF PDOs (EFFICACY)

Assessment of Achievement of Each Objective/Outcome

30. **The project experienced a very modest progress during the first four years of implementation. After subsequent restructurings it reached a substantial achievement of most outcomes though overall performance remained uneven.** The first restructuring was approved after an implementation period of more than four years that had achieved very modest or negligible progress. The project's achievement of the PDO indicators ranged from 0.0 to 32,2 percent, or 0.00 to 78 percent, depending on the date of comparison. Subsequently, upon the revision of targets agreed through two additional restructurings, the project reached at closing a higher achievement of the outcomes, i.e. between 84,8 to 99,7 percent for three of the four main PDO indicators (Table 3). The causes of the project's poor performance are discussed in Section III.



31. A split rating method is applicable when outcome targets have been formally revised by a restructuring process. Its application is a requirement in the case that the scope of a project has been narrowed, which is the case of this ICR, as all end targets were reduced. Furthermore, the marked contrast of the project's performance between the periods before and after restructuring warrants the application of a split rating method in securing a balanced assessment. Such method measures the project's achievements against the targets established for the period prior and after restructuring, thus, on one side, internalizing the improvements achieved through the 'corrections' made by the restructuring while also taking into account the impact of an early period of poor performance (i.e. it seeks a balance acknowledging the benefits of the revision while mitigating the possible distortion of an adjustment that 'lowers the bar').

32. The split rating method was applied considering two periods: the initial four-year period between project approval and the first restructuring in mid-2020, and a second period from mid-2020 till closing date. The assessment yielded the following efficacy ratings: Negligible and Modest-to-Substantial, respectively. The end targets and actual achievement of PDO/outcomes for the main –and breakdown/supplemental– indicators, as well as for both periods are presented in Table 3. Details on the application of the split rating, including assumptions, justification of the approach and breakdown of ratings are presented in Annex 5.

Table 3. PDO Indicators Achievement – Prior and After Restructurings

PDO Indicator	Period 1: 2016-mid2020				Period 2: mid2020-end2023		
	End Target	actual by 2020	achieved % by 2020	achieved % by 2023	End Target	actual by 2023	achieved %
Direct project beneficiaries	765,000	74,533	9.7	25.4	229,000	194,302	84.8
Direct project beneficiaries that obtained enhanced access	241,000	3,386	1.4	4.4	24,900	10,608	42.6
Direct project beneficiaries that obtained access	524,000	71,147	13.6	35.1	204,100	183,694	90.0
Female beneficiaries (%)	40	32	80.0	100.0	40	40	100.0
People provided with access to electricity by household connections	175,000	56,364	32.2	78.1	139,600	136,748	98.0
People provided with new elec. by household connection–Only renewable	173,000	56,364	32.6	79.0	139,600	136,748	98.0
People provided with new elec. by household connection–Other than renewable	2,000	0	0.0	0.0	—	—	—
Community electricity connections constructed under the project	1,800	105	5.8	17.7	780	319	40.9
Community elec. connections constructed–Offgrid/minigrid–Only renewable	1,750	105	6.0	18.2	780	319	40.9
People provided with new elec. by household connection–Other than renewable	50	0	0.0	0.0	—	—	—
People that obtain daily and constant electricity service through mini-grids.	37,000	0	0.0	0.9	340	339	99.7
People that obtain a daily and constant electricity service and with prior access	28,000	0	0.0	1.0	290	288	99.3
People that obtain a daily and constant electricity service without prior access	9,000	0	0.0	0.6	50	51	102.0

Justification of Overall Efficacy Rating

33. No single overall rating is assigned, as indicated above the application of the split method yields ratings ranging from Negligible to Substantial. This range of ratings is incorporated into the weighted average assessment aimed at defining the project's Overall Outcome Rating.

C. EFFICIENCY

Assessment of Efficiency and Rating

34. An ex-post analysis of the Project aimed at replicating the PAD's gives the following results: an economic internal rate of return (EIRR) of 21.3 percent and its net present value (NPV) US\$18.2 million. These positive results



compare to an EIRR of 28.8 percent estimated at appraisal for an equivalent component, and a NPV of US\$59.6 million. The lower economic indicators achieved by the project are attributed to, both, higher capital costs per unit of equipment and the considerable reduction in the size of the investment component of the project (details in Annex 6). Considering that there is some degree of uncertainty around the 3G SHS equipment's economic life, a sensitivity analysis was undertaken for a shorter lifetime -from 10 to 6 years. It was found that in such case the EIRR would go down to 17.9 percent. Also, if the SHS 3G is excluded, the project's EIRR would decrease to 13.7%, thus confirming that the incorporation of a significant SHS 3G component had a positive impact in the project's economic performance.

35. **The results of the economic analysis should be considered conservative, since there are important unquantifiable benefits that are not included in the assessment**, such as those arising from improvements in the quality of life of beneficiaries (in terms of education, safety, leisure time) and the productive activities (income increase, employment generation, both associated to the country's long term objectives) made possible by the access to an enhanced electricity service. Hence, it should be acknowledged that the economic analysis does not embody a full assessment of the project's economic benefits.

36. **The Project's design and implementation paid special attention to an efficient use of resources. To this end, it relied on the PCU (within the Sub-Secretaría de Energía Eléctrica-SSEE) that had been working efficiently on a previous WB project.** Taking advantage of the existing experience, the PCU took a centralized role in the implementation of the project, including procurement activities in close coordination with the PEUs, as well as in financial management and in assuming overall responsibility on the Project's Monitoring and Evaluation (M&E).

37. **In addition to utilizing the capacity built in previous projects, the Project's design incorporated a set of capacity building activities as part of Component 4 aimed at strengthening the operating capacity of the PCU and the PEUs.** Nevertheless, implementation had a slow start, especially during the first two years, as loan effectiveness took longer than expected (mainly due to disruptions associated to a change in administration and sector authorities) and procurement processes were slow, particularly in the evaluation and selection of contractors.

38. **In mid-2020 the project implementation role was split into two units –technical and fiduciary– as a PCU/Fiduciary (DiGePPSE – Dirección General de Programas y Proyectos Sectoriales y Especiales) was established within the Secretaría Legal y Administrativa (Subsecretaría de Administración y Normalización Patrimonial),** to assume responsibility for administrative, budget and financial aspects, including fiduciary, procurement processes and legal aspects. Although the separation of roles was rationale from a management viewpoint, and both units had the required capacity, coordination proved to be difficult, particularly during the initial transition period. At the same time, the project was moved from the Ministry of Productive Development to the Ministry of Economy.

39. Taking into account the Project's positive though lower than expected economic performance and its relatively slow implementation, it is rated **Substantial to Modest** for efficiency.

D. JUSTIFICATION OF OVERALL OUTCOME RATING

40. **A Moderately Unsatisfactory** rating is assigned for the project's overall outcome based on the application of the split rating method, which considers the significant shortcomings of the operation in partially achieving the PDO, particularly due to its poor performance during the first four years of implementation, and the significant reduction of its end targets, in spite of the improved performance during the last year, the high relevance of the project's objective and its substantial to modest efficiency. The application of the split rating is presented in Annex 5.



E. OTHER OUTCOMES AND IMPACTS (IF ANY)

Gender

41. The Bank's safeguard policies do not provide for a Gender Action Plan that would have identified gender gaps and measures to address them, however, the results framework included one gender indicator: a 40 percent target for female beneficiaries. This target was met, that is, at least 40 percent of the project's beneficiaries were women (around 78,000). Furthermore, a toolkit for preventing sexual abuse was applied yielding satisfactory results (Section IV B). Surveys undertaken during the last two years of implementation focused on project satisfaction and concluded that the project had a positive impact on women. The provision or enhancement of electricity services improved the household's living conditions, in particular for women since electricity facilitates multiple tasks (both productive and reproductive) and offers additional time for other tasks (education, paid work, leisure). Mini-grids provided the prospect to develop alternative economic activities, such as tourism, thus enhancing women's opportunity to engage in productive activities. Also, the provision of electric fences for livestock benefited mostly women since they usually assume this duty in rural areas.

Poverty Reduction and Shared Prosperity

42. The main beneficiaries of the project included some of the poorest of Argentina, since poverty rates are higher in rural areas, including a large component of indigenous peoples (34 percent of the communities provided with electricity connections corresponded to indigenous peoples). The provision of a modern electricity service through the implementation of mini-grids and isolated systems, will have a positive impact on reducing poverty and improving the quality of life of around 48,000 rural households, as they will reduce considerably the use of traditional and less effective energy resources, and will potentially improve the quality of health and education, better security, and the opportunity to engage in productive activities that would not have been possible otherwise.

III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

43. The project was aligned with the [CPS 2015-18 for Argentina](#) and the universal access goals of the GoA. Project preparation was undertaken within a positive context characterized by the following main factors:

- (a) **The GoA had a strong ownership of rural energy projects.** The project was consistent with the GoA's main goals of reducing extreme poverty and fostering social inclusion, thus providing a strong political rationale for the Government to continue its solid support of PERMER II. In fact, regardless of changing administrations, the project was perceived as the main national instrument to move Argentina toward universal access, while also contributing in diversifying the energy matrix and reducing environmental impacts. Nevertheless, it is noted that being access a relatively minor issue (economically and politically) compared to other energy sector problems, it competes unfavorably for the Energy Secretariat's attention.
- (b) **The Bank-financed PERMER I had been the most successful rural electrification project implemented by the GoA in previous years.** PERMER I had proven to be a valuable mechanism to increase electricity access in rural areas in a sustainable manner that had created also a rural network comprising federal and provincial institutions, bridging national and subnational entities. Hence, the project relied on the capacity built during the implementation of PERMER I, including an experienced PCU and similar implementation arrangements.
- (c) **An overall optimistic perception that risks were moderate or low.** While macroeconomic imbalances had



increased over the years prior to project approval, and Argentina was experiencing a mild contraction, the potential impact of macroeconomic risk was considered to be moderate. Also, while it was acknowledged that political events and administration cycles could negatively impact project progress, this impact was deemed moderate. Finally, it was considered that the project would not be affected by the existing sector imbalances caused by fuel imports and related subsidies. In sum, the assessment undertaken during project preparation identified correctly the main risks that could impact project implementation and sustainability but was inclined to underestimate their magnitude.

B. KEY FACTORS DURING IMPLEMENTATION

44. **The project had a slow start caused by its late effectiveness and, subsequently, difficulties in gaining momentum typical of an operation that involves the participation of a large number of stakeholders.** Twenty-one Participation Agreements had to be signed between the Federal Government and each participating Province for the project to become fully operational. Also, agreements were reached with other parties that collaborated in different rural aspects, such as the INTA (Instituto Nacional de Tecnología Agropecuaria). In sum, the first years (till 2018) were spent mostly in preparation, completing institutional arrangements, design of technical specifications and procurement processes.

45. **Main factors that affected project's execution during either the entire implementation, or a specific period, were:**

- (a) **Continuous changes on the organization of the energy sector, as well as on authorities, were a main factor of delay.** Multiple sector structural changes, including several locations of the Secretariat of Energy within the GoA, and the associated changes in authorities proved to be highly disruptive to the continuity of activities. Selection of contractors, signing of contracts and payments were delayed by multiple disruptions and transitions. Though to a lesser degree, changes in the project's team also prevented a smooth implementation during specific periods
- (b) **Import restrictions issued by the Central Bank (BNA, Banco de la Nación Argentina) in March 2022 in response to growing macroeconomic imbalances, had a direct impact on the execution of a large number of subprojects.** Contractors experienced serious difficulties in importing goods, such as solar panels and batteries, that hampered the execution of ongoing subprojects. Consequently, at project closing, 34 contracts could not be completed. It was therefore agreed that unfinished contracts would be concluded and financed under a new IBRD operation³. The 34 contracts have been amended to be considered part of the new IBRD operation. The new administration is assessing the status of each contract as well as the budget constraints and is undertaking the needed due diligence to assure the continuation of all contracts that have initiated implementation.
- (c) **The COVID-19 pandemic, an exogenous and unforeseen event, was a factor that slowed down most activities.** Public health measures taken by the GoA –including social distancing policies and a strict lockdown– and suppliers' governments prevented normal activities in the field as well as the timely production and delivery of goods. Also, part of the loan was cancelled with view to redirect resources for the containment of the pandemic.

46. Table 4 presents a more comprehensive set of the factors affecting project implementation and the main actions taken by different players, categorized in three groups, depending on the level of control that different players had on each factor. Annex 4 presents a timeline of the project that identifies a set of mostly external events under which the project was implemented.

³ Clean Energy for Vulnerable Households and Communities Project (P178553), declared effective on October 2023.



Table 4. Factors That Affected Project Implementation

Key factor	Subject to control of government entities responsible for the project	Subject to World Bank control	Outside the control of the project or the energy sector
Major delays in Project Effectiveness and Implementation during initial four years.	• Late project effectiveness upon delays in the signing of the first two Participation Agreements between SE and participant provinces (effectiveness was reached 7 months after loan signing and 13 months after approval). • Changes in project staff in early 2017, including PCU coordinator, caused delays on design, specifications and procurement. • Temporary delays in defining a measuring methodology (M&E) and in establishing the GRM. • Procurement arrangement that involved a large number of members in the evaluation committees constituted a cause of delay that was never entirely overcome. Also, procurement was delayed by unnecessary and/or redundant administrative steps. • Action Plan, agreed with WB, to address main implementation problems.	• Bank carried out the MTR (May 2019), to assess the achievement of PDO, implementation pace and to map ways forward. • Action Plan, agreed with GoA, to address the main implementation problems.	• 2015: change in administration, including sector authorities, also delayed effectiveness. • In 2018 the Ministry of Energy was absorbed by the Ministry of Finance. While energy access policy was not significantly changed, several processes were delayed. Consequently, the project was not able to document payments during a period of 5 months. • 2019: changes in administration and sector authorities delayed the first restructuring. • Government wide objective to reduce fiscal expenditures posed obstacles on the recruitment of specialized technical professionals for the project. • Transition in administrative procedures caused by new (2016) principles for public transparency.
	• The project's first restructuring agreed between the GoA and the Bank in 2020, updated the Results Framework and project schedule (closing date extended by 20 months), and cancelled part of the loan, among other changes.		



Key factor	Subject to control of government entities responsible for the project	Subject to World Bank control	Outside the control of the project or the energy sector
COVID-19 pandemic caused delays in Argentina as well as in suppliers' countries	Emergency measures given by the Federal Government as a health protection policy slowed down significantly most project activities during great part of 2020 and 2021.		- Governments of suppliers' countries also took health protection measures that delayed production and trade activities. - GoA decided to redirect IBRD and other government resources to public health activities aimed at the containment of the COVID-19 pandemic
	Joint GoA/Bank response: first restructuring of 2020 extended the loan's closing date to compensate for delays caused by the pandemic and other factors, and cancelled partially the loan (by US\$31,3 million) with view to redirect these resources to public health activities.		
Macroeconomic context – severe import restrictions caused significant delays on most contracts executed during the last 2 years of implementation.	- Project contractors experienced serious difficulties in importing goods, such as solar panels and batteries, that hampered the execution of subprojects. - Upon the request of contractors', the SE and the Ministry of Economy were able to facilitate and expedite some imports under the project but, overall, implementation and disbursements kept lagging. - In spite of two additional loan extensions, for 18 additional months, 34 contracts could not be completed on time.		In March 2022 the BNA issued regulations that severely restricted imports in response to overstocks experienced during 2021. New regulations on this topic, issued in August 2022, complicated further the import process.
Project Restructurings	Project restructurings were the main GoA/Bank management instrument used in addressing implementation problems and in adapting to changing conditions. The second restructure (June 2022) proceeded to cancel Component 2: Solar thermal energy (due to almost negligible progress and human resources constraints the SE decided to focus its efforts in moving the electrification component) and reallocate its resources among other components. Section I B: Significant Changes During Implementation, elaborates on the objectives and content of the three restructurings.		



IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

M&E Design

47. **Overall, the set of indicators reflected well the project's objective and the outputs expected from each component, that is, the set is consistent with the Project's theory of change.** The M&E design reflected the scope and objectives of the project that relied on a large scale and wide-ranging Component 1 (Renewable electricity service provision) plus the outcomes of two other components. The Results Framework included four main PDO indicators (plus nine supporting PDO indicators: eight breakdown and one supplemental) and a large number (20) of intermediate results indicators. The indicators were measurable though the targets originally established for a few indicators (such as the number of community electricity connections) appeared to be out of scale or the result of a misinterpretation. Also, the unusually large number of indicators (33 in total) may have posed an excessive monitoring effort.

M&E Implementation

48. **M&E during implementation was satisfactory, it was handled effectively by an experienced PCU and adapted to changing conditions.** Responsibility for M&E relied on the PCU, that had experience in handling this function in previous Bank operations. Under its guidance, the PEUs collected data on subprojects and from beneficiaries. To ensure an effective M&E, the project provided capacity building in this area for the PCU and other stakeholders. The Operational Manual (OM) established the responsibilities of all parties in regard to M&E. Accordingly, the PCU issued bi-annual reports that comprised a detailed update of the project's implementation progress, including an update of the results framework indicators, as well as the monitoring of sub-projects' progress, contracts and safeguards compliance. This progress was regularly reflected in 16 Implementation Status and Results Reports (ISRs) produced by the Bank team during implementation.

49. **M&E was rated moderately satisfactory through the whole implementation with the exception of a short period, in mid-2018,** when it was downgraded to marginally unsatisfactory after a prolonged delay in defining the required measuring methodology.

50. **M&E implementation was flexible. Each of the project restructurings included a change in the results framework in order to adapt it to new conditions and more realistic outcome expectations.** While PDO indicators remained the same, the number of intermediate indicators was reduced considerably, nine of the original twenty indicators were dropped (including those linked to the cancelled component 2) and a new one was added. Also, end targets that appeared out of scale were corrected.

M&E Utilization

51. **M&E data was regularly reflected in 16 ISRs produced by the Bank team during implementation.** These reports were used by the Bank in identifying –and informing management on–the potential implementation challenges in various areas and in monitoring closely the project's slow progress during initial years, resource constraints, end-of-



implementation challenges and the need for adjustments to achieve a satisfactory closing. An efficient use of the M&E framework made possible a timely and effective restructuring of the project in three occasions.

Justification of Overall Rating of Quality of M&E

52. Given its adequate design, effective implementation and utilization in key decisions, but some moderate shortcomings of design and initial implementation, the quality of the M&E is rated **Substantial**.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

Social and Environmental Safeguards

53. **The Project triggered six World Bank safeguard policies as identified in the PAD.** These were: Environmental Assessment, Natural Habitats, Indigenous Peoples, Physical Cultural Resources, Involuntary Resettlement and Projects on International Waterways. The Project's safeguards category was set as B. It was considered to have limited and reversible environmental impacts that could be readily mitigated.

54. **Given that the subproject's potential environmental and social impacts could not be identified prior to project approval, an Environmental and Social Management Framework (ESMF) was adopted.** This ESMF was based on a revised and updated version of the previous framework prepared under PERMER I, and was subject to consultations with local and provincial governments, local nongovernmental organization and stakeholders' representatives, including INAI (Instituto Nacional de Asuntos Indígenas). The PCU's staff included both social and environmental specialists that monitored works in the field.

55. **Environmental safeguards were implemented without significant problems. While the diversity of sub-projects required closed monitoring, the interventions across the project components did not produce any major environmental impact.** The ESMF included a screening process for sub-projects and Environmental Impact Assessments (EIAs)/ Environmental Management Plans (EMPs) were consulted upon and disclosed prior to the start of works. Accordingly, the overall safeguards performance was rated Moderately Satisfactory throughout the entire implementation period. Cases that warranted rating downgrades (though minor) on specific safeguards were:

- Compliance with safeguard OP 4.04: Natural Habitats, was downgraded to Moderately Satisfactory in December 2023 (ISR16) due to an impact recorded in the Luracatao site caused by the extraction of a legally protected cactus and the absence of clear measures to avoid this to happen.
- Compliance with safeguard OP 4.11: Physical Cultural Resources, was downgraded to Moderately Satisfactory in December 2023 due to major damages on a hermitage at Luracatao. Remediation was still pending at project closing.

56. **The Project triggered on Indigenous Peoples and Involuntary Resettlement safeguards. Consequently, an Indigenous Peoples Planning Framework (IPPF) and a Resettlement Policy Framework (RPF) were prepared by the SSEE,** based on the experience of PERMER I and other relevant sources. Both frameworks were subject to consultation and were subsequently disclosed in the country and the Bank's web. The IPPF defined procedures and principles to ensure a process of free and informed consultation with the affected Indigenous Peoples' communities.

57. **While the participation of indigenous peoples in the sub-projects preparation and implementation, as well as the application of RPF, required close monitoring to ensure their compliance with safeguards, the outcome was, overall, satisfactory.** Delays persisted in achieving cultural adequacy. Safeguard OP 4.12 was rated satisfactory during great part



of the implementation period, but it was downgraded to moderately satisfactory in May 2022 as it was revealed that some RPF procedures related to micro-grids were implemented by local teams with uneven capacity. One Resettlement Action Plan was prepared for the Luracatao micro-grid to address one case of involuntary acquisition and 42 cases of economic displacement. At project closing, the resolution of 19 of these cases was still pending. It has been agreed that these cases will be monitored and addressed under the ongoing Clean Energy project (P178553).

58. **The Bank's supervision team paid special attention to sexual exploitation and abuse risks.** Project activities did not imply the presence of many workers, and the risk was rated to be low, however, tender documents included recommendations to give preference to hiring indigenous community local workers, to include local facilitators to ensure cultural adequacy and to avoid potential impacts of gender-based violence.

59. **It is reported that the sustained application of World Bank social safeguards in the country is contributing toward a greater awareness on the subject and, in particular, a greater respect toward the rights of indigenous peoples.** This contribution in local culture is recognized by both specialists working in the field and beneficiaries. Indigenous Peoples Plans were prepared for mini-grid sub-projects and user's manuals and related communications were translated into three indigenous languages.

60. **A Grievance Redress Mechanism (GRM) was established in mid-2018 to provide remedies to parties who perceive that their rights had been adversely affected by the Project.** To that end, pertinent documents were published and the SE (with some delays in updating its website) set up a call center to collect claims. The PCU produced culturally-adapted communications materials. Also, provisions for the proper application of the GRM were incorporated into bidding documents. There is evidence that the PCU processed complaints and comments received directly or indirectly (through participating provinces and/or contractors). Repeated reminders of the Bank for the PCU to register and provide full reports on the complaints received and how were they addressed, as well as the reminder that GRM applies to all aspects of the project, suggest that there were shortcomings in its implementation.

Financial Management and Audits.

61. **A financial management (FM) assessment for the proposed project arrangements was conducted at the time of appraisal.** These considered the centralization of project administration in the hands of the PCU and the General Financial Management Directorate (DGCAF), both within the internal organization of the SE. In practice, in mid-2020 the responsibility of project implementation was split into two units –technical and fiduciary– as a fiduciary unit (DiGePPSE) within the Secretaría Legal y Administrativa (Ministry of Economy) assumed responsibility for administrative, budget and financial aspects, including fiduciary and legal aspects (as well as procurement). While the separation of roles was rationale from a management viewpoint, and both units had the required capacity, coordination brought a new challenge, particularly during the initial transition period.

62. **Centralization of project administration in the hands of qualified and experienced staff led to satisfactory FM during implementation. Measures to strengthen financial management were implemented with the support of project proceeds** (Component 4). The Project consistently provided timely and reliable financial information for monitoring purposes. External audits did not identify qualified opinions and were presented in a timely fashion, with the exception of the 2019 audit which was delayed by the pandemic and, subsequently, the Bank extended its deadline. Financial monitoring reports were delivered in a timely fashion and recommendations were implemented on an ongoing basis.

63. **In addition to its regular supervision missions, that often reported on the subject, the Bank launched in 2023**



a mission to address specifically FM issues. The mission concluded that financial management arrangements (FM staffing, budgeting, accounting, internal control funds flows, financial reporting and auditing) adhered to the standards required by the World Bank. On the whole, FM issues required only moderate supervision and Bank inputs. The overall project FM risk rating was maintained Moderate through the entire implementation period as there were no significant FM risk factors.

64. **Legal Covenants.** The loan agreement included three legal covenants related to FM issues: the signing of Participation Agreements and Operation, Maintenance and Replacements Agreements, and keeping a Specific Budget Line Entry in the national budget. All covenants were complied with.

65. FM was rated **Satisfactory** through all the implementation period as well as at project closing.

Procurement

66. **Procurement for the project was carried out in accordance with the provisions stipulated in the Loan Agreement:** the World Bank “Guidelines: Procurement of Goods, Works and Non-consulting Services under IBRD Loans and IDA Credits and Grants by World Bank Borrowers” issued in January 2011 and revised in July 2014, and the “Guidelines: Selection and Employment of Consultants under IBRD Loans and IDA Credits and Grants by World Bank Borrowers” issued January 2011 and revised in July 2014.

67. **The structure for carrying out procurement activities, monitoring and supervising the fiduciary arrangements was defined by the capacity assessment undertaken during the preparation of the project.** It was confirmed that the PCU had the overall capacity to carry out such role. In practice, such role was complemented by the DiGePPSE, which took a leading role in the procurement process until the award of contracts. All procurement activities were coordinated with the beneficiary provinces which participated -to greater or lesser degree- in providing information, the preparation of bidding documents and the evaluation of bids. The project provided resources (Component 4) to strengthen the operational capacity of the PCU and PEUs, including procurement activities.

68. **Procurement processes constituted one of the main factors of delay which made necessary several extensions of the loan’s closing date.** In spite of its centralized approach and the proven capacity of the PCU and DiGePPSE, the involvement –and coordination with– more than twenty participant provinces posed a difficult challenge due to the differing conditions of beneficiaries, the capacity building requirements for a large number of people involved in the process, and the need to coordinate a large number of parallel activities. Ninety contracts were signed and executed as part of Component 1. Besides the external factors discussed in Section III (i.e. impact of the pandemic and severe import restrictions) problems or factors of delay reported during different phases of implementation were:

- (a) Need to improve the planning of procurement activities, in particular in cases that involve different processes that are connected to each other and require for documents to be prepared and launched in parallel.
- (b) Delays in the evaluation, selection and awards process – slow preparation of bidding documents, delays in the approval of award recommendations and contract signing.
- (c) Lack of concurrent procedures to identify and prevent bottlenecks.
- (d) Changes in authorities or in the organization of the sector caused delays in approving specific steps or the signing of contracts.
- (e) Other factors were: the need for more accurate or realistic cost estimates, temporary shortages of specialized staff, and an excessive number of members in evaluation committees.



69. **The experience in contracting solar water pumping facilities illustrates some of the procurement problems faced by the project.** The project's productive uses sub-component included water pumping facilities aimed to benefit 3,000 small agricultural producers in ten provinces. The bidding process started in April 2021 but the evaluation of bids and award took more than one year. Sluggish progress was caused by an underestimation of costs, slow evaluation process, and delays in the approval for the award and signing of contracts. By the time contracts were ready to be signed contractors opted to refrain to do so since, given the impact of inflation and the extended period, original bids were no longer viable financially. Consequently, the US\$6.4 million subcomponent was cancelled.

70. **The project's procurement performance constituted an important challenge throughout the entire implementation period. Bank ratings on procurement performance fluctuated from Moderately Satisfactory to Moderately Unsatisfactory, closing in MU.** While all contracts were signed, at the end a significant number of contracts (34) were not completed at project closing. It was therefore proposed that these contracts will be concluded and financed under the new Bank operation (Clean Energy Project). PERMER's procurement experience warrants a systematic review that could be particularly useful in identifying areas to improve and draw lessons for future operations.

C. BANK PERFORMANCE

Quality at Entry

Rating: Satisfactory

71. **Project design involved a sound assessment and proposal of an approach that had been successful in the PERMER I, including its implementation arrangements and the use of known and proven technologies. The PDO was consistent with the scope and objectives of the CPS and remained relevant throughout the whole implementation period.** Accordingly, it incorporated a set of cross-cutting initiatives introduced by the CPS, such as the provision of knowledge services, in-depth assessment for impact evaluation and a greater focus on low-income areas. Also, the project's institutional arrangements were designed with view to strengthen the rural network comprising federal and provincial institutions that had been established by PERMER I. That is, project design followed a programmatic approach suitable to the long term nature of the rural electrification effort.

72. **Sound preparatory work was characterized by a comprehensive assessment of all aspects, including safeguards, the project's economic viability, confirmation of the executing agency's implementing capacity and fiduciary arrangements, and the incorporation of lessons learned in previous projects.** The PCU was based on the implementing unit of a previous successful rural electrification project. Nevertheless, Component 4 provided for additional resources to strengthen the operational capacity of the PCU and participating provinces and their respective PEUs to carry out the activities under their responsibility. It is noted, however, that while project risks were identified correctly during the preparation phase, these were underestimated.

73. **The design of the M&E framework was consistent with the project's objective and the outputs expected from each component.** However, while the Results Framework included a set of relevant and measurable indicators, the end targets originally established for some indicators seemed to be out of scale or the result of a misinterpretation. Also, the large number of indicators (33 in total, including PDO and intermediate indicators) appeared to pose an excessive monitoring task. These shortcomings were corrected during implementation.

Quality of Supervision

Rating: Highly Satisfactory



74. **The Bank actively supervised the project, the frequency and composition of missions was adequate (two per year as an average), in spite of the constraints posed by the pandemic.** The Bank engaged with authorities from the early start, keeping the focus always on the achievement of the PDO and the supervision of fiduciary aspects and safeguards.

75. **Reporting on project implementation was objective, timely and consistent throughout the entire period.** The team issued fifteen aide memoire and sixteen ISRs that reported on the main implementation issues drawing attention to the main challenges, and monitored effectively the project's progress vis-à-vis the results framework. Performance ratings were objective and justified.

76. **Supervision of environmental and social safeguards and fiduciary aspects was satisfactory,** both in terms of the quantity of resources allocated and quality of supervision, including numerous field visits. It is worth noting the counterpart's acknowledgment of the valuable and timely support provided by the Bank's local staff in addressing all safeguard and fiduciary aspects of the project.

77. **The Bank's dialogue with the Borrower and related agencies was constructive and useful in overcoming implementation obstacles.** The Bank was responsive and flexible in dealing with most implementation problems. Upon the Government's request to restructure the project in three occasions the Bank responded in a practical and timely fashion. All restructurings entailed amendments aimed at adapting to changing conditions –delays caused by factors internal and external to the project, the need to redirect resources to priority public health needs, and the reallocation of resources among components– and, hence, were justified. It is noted that the first restructuring was late, however, the causes of this delay (changes in administration and sector authorities, and the pandemic) were exogenous to the Bank and the project's management. Possibly the only shortcoming was the Bank's disagreement on longer loan extensions which could have enhanced restructuring benefits.

Justification of Overall Rating of Bank Performance

78. Considering the preparation and supervision ratings, an overall rating of **Satisfactory** is assigned.

D. RISK TO DEVELOPMENT OUTCOME

79. **The project replicated the successful institutional arrangement of its predecessor (PERMER I) with view to guarantee its operational sustainability.** The project's development outcome is to provide access to modern energy services in selected rural areas. As presented in Section II B, after three restructurings (and the explicit reduction of end-targets) the project succeeded in meeting a significant level of the expected outcomes, thus providing energy services to 194,000 people. While PV technology entails some degree of complexity its sustainability relies mostly on institutional - and logistical- rather than on technical factors, as the main challenge is to provide effective maintenance and customer service in a widely dispersed market and, often, under difficult geographical conditions. The project was designed taking into account the successful experience of PERMER I which established a robust institutional arrangement to guarantee an effective Operation&Maintenance&Replacement (O&M&R) through agreements with each province that were incorporated into the OM, and the participation of quality service providers. Such arrangement was incorporated into PERMER II's design. Also, the emphasis on mini-grids was somehow fostered by the strong O&M scheme for these technologies as local utilities are fully in charge of their operation. Furthermore, the approval of the follow-up Clean Energy project provides the resources to support the project's sustainability. However, the introduction of a third-generation solar technology (3G) for households, that is not fully supported by such institutional arrangement, casts doubts on the sustainability of this sub-component.



80. **Project risks were higher than expected at appraisal. Uncertainties associated to macroeconomic and political factors, including the GoA's commitment to rural energy, could affect the project's sustainability.** The overall risk rating of the project at appraisal was considered to be Moderate, without any specific risk being rated at a higher level. Supervision records reveal a different picture as risks were higher than expected and the project closed with an overall risk rating of Substantial. Two main areas of high risk that impacted negatively on the project's implementation, and may still affect its sustainability, are the country's macroeconomic situation and some degree of political instability. The recent change of administration and overall policy orientation that appears not to be inclined to support rural energy operations brings about a new source of uncertainty that may impact on the project's sustainability.

V. LESSONS AND RECOMMENDATIONS

81. The assessment of the project's preparation and implementation experience leads to the following broader lessons and recommendations:

- (a) **The project's experience confirmed the importance of strong government ownership and previous effective operations in the success of a project.** However, these positive factors need to be complemented by a rural electrification policy to be formulated, and implemented, within a rural development context. Upon the absence of a proper sector policy, Bank supported PERMER operations filled the gap during the last decades and were even perceived as the country's access program. The absence of such policy could weaken government ownership and compromise the sustainability of the project's long-term objectives. Also, an energy access policy should be designed within a broad rural development context –i.e. in coordination with other sectors of rural interest, such as housing, water, health, education, natural habitat– to ensure complementarities and enhance the effort's efficiency.
- (b) **The impact of organizational changes and/or the replacement of authorities on the implementation of a project is often much higher than expected and often becomes a major factor of disruption and delays.** Multiple changes in the sector's organization, including several locations of the SE within the GoA, and the associated changes in authorities and processes, proved to be highly disruptive to the continuity of PERMER's activities and a main factor of delay. Transition costs inherent to organizational changes cannot be underestimated and should be incorporated by governments into the decision process and, to the extent possible, into the dialogue with the Bank.
- (c) **Providing access to electricity in rural areas is, by nature, a complex long-term effort which is suitable for a programmatic approach. The project's approach in relying on the experience of the predecessor PERMER I project proved to be highly useful both during project preparation and its implementation.** Main lessons and/or experiences adopted by the project were the need to continue strengthening the rural network engaging federal and provincial institutions, reliance on the capacity built by PERMER I (including the existing PCU) and replicating or improving implementation arrangements for social and environmental matters (ESMF and RPF experience) and a robust institutional arrangement for O&M&R, as well as the use of proven technologies.
- (d) **Regardless how thorough is the preparation and design of a project, it will be subject always to unforeseen and external factors that can impede its successful implementation. It is therefore essential to incorporate into the project's design an effective protection against unforeseen events.** PERMER II was negatively affected by two factors external to the project control: the COVID-19 pandemic and severe import restrictions established by the BNA. Options to incorporate adaptability are the provision of a sizeable unallocated budget to address unforeseen needs (such as CERC), legal agreements open to –or that allow a smooth– restructuring of the operation, as well as the flexible use of management tools that can be updated when necessary (e.g. Operational Manuals). As PERMER's experience confirmed, incorporating this flexibility into the dialogue with



the Bank was critical in addressing effectively the client's needs and changing preferences during implementation.

- (e) **The incorporation of a component aimed at promoting the productive use of electricity enhances the social and economic benefits of rural electrification projects. Key factors contributing toward a successful productive uses effort are the participation of entities with local rural knowledge in the preparation and implementation of the operation, and the need to align to broader rural development approach.** In particular, PERMER's experience in signing a cooperation agreement with INTA, a technical institution with extensive experience in agriculture and livestock, proved to be highly useful in providing capacity and a better understanding of rural development needs thus ensuring the successful implementation of the productive uses sub-component (particularly electric fencing). Also, the promotion and implementation of productive uses of electricity in rural areas implies the alignment to a broader rural development scope (since it will benefit –or interact with– others sectors, such agriculture, livestock, water supply, agro-industry) and, hence, the necessity to overcome common limitations in the coordination between sectors.
- (f) **Procurement activities of rural energy project pose a significant implementation challenge due to multiple factors, such as differing regional needs and conditions, significant capacity building requirements and the large number of parallel activities. PERMER's complex experience warrants an ex-post assessment to draw lessons for future operations.** It is recommended to assess systematically the project's procurement experience in all its phases, from its implementation arrangement and administrative process to the efficiency achieved in each step, focusing on the identification of obstacles and factors of delay, the potential to cut redundant or unnecessary steps and the need for improved planning and monitoring mechanisms to anticipate and mitigate problems. The findings of such assessment could be potentially useful for future Bank operations that are characterized by complex procurement processes implying a diverse and large number of activities.
- (g) **Establishing a strong PCU is a proven approach for the successful implementation of a project, however, it is a short-term solution that may hinder the country's long term capacity building objectives.** As in PERMER's experience, PCU's tend to be an isolated entity characterized by an above average capacity that fulfills its project implementation role. However, given their 'external' condition (often staffed by specialized medium-to-short consultants) its capacity and valuable experience seldom contributes towards building capacity in a broader sense. Strengthening public institutions in a sustainable manner requires mainstreaming the knowledge gained through World Bank operations. The conflict between the PCU model and capacity building objectives has yet to be resolved. It is recommended that the design of projects should include a more forceful intent to build capacity and explore options to achieve this goal.
- (h) Stakeholders specific proposals for improving the implementation and sustainability of rural energy projects include;
 - (i) **Add flexibility into turn-key contracts with view to allow for the partial certification of works in case some components cannot be delivered due to factors of delay beyond the control of the parties.** In PERMER, the impact of the severe import restrictions went beyond the overall delay of sub-projects, since it impeded the payment for part of the works –that did not depend on imports– that were actually completed, due to the absence of a partial certification provision in the contracts.
 - (ii) **Since the technology for rural electrification keeps evolving it is necessary to incorporate flexibility into the selection of technological options,** i.e. agreements and operational manuals should be open to select new technologies that offer the potential of more efficient and cost effective solutions.
 - (iii) **Establish a platform for the integrated management of all project contracts,** with real-time interactive access of all interested parties in order to monitor and update the progress of all contracts, identify potential problems and the required measures in a timely fashion.



ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS INDICATORS

A.1 PDO Indicators

Objective/Outcome: Direct project beneficiaries

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Direct project beneficiaries	Number	0.00 01-May-2015	765,000.00 09-Oct-2015	229,000.00 12-Jun-2023	194,302.00 31-Dec-2023
Direct project beneficiaries that obtained enhanced access (daily hours and quality of service) to modern energy services (electricity, water pumping, productive uses, solar thermal applications)	Number	0.00 01-May-2015	241,000.00 09-Oct-2015	24,900.00 12-Jun-2023	10,608.00 31-Dec-2023
Direct project beneficiaries that obtained access to modern energy services	Number	0.00 01-May-2015	524,000.00 09-Oct-2015	204,100.00 12-Jun-2023	183,694.00 31-Dec-2023



(electricity, water pumping, productive uses, solar thermal applications)					
Female beneficiaries (%)	Percentage	0.00	40.00	40.00	40.00
Comments (achievements against targets): Achievement of the main beneficiaries target, i.e. total beneficiaries, was of 84,8 percent, including 88,2 percent for beneficiaries that obtained a new service, as the project focused in this objective. In contrast, achievement of the enhanced service target was 40 percent. The expectation of female beneficiaries was fully met.					

Objective/Outcome: People provided with access to electricity by household connections

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
People provided with access to electricity by household connections	Number	0.00 01-May-2015	175,000.00 09-Oct-2015	139,600.00 12-Jun-2023	136,748.00 31-Dec-2023
People provided with new elec. by household connection.– Offgrid/minigrid–Only renewable sources	Number	0.00 01-May-2015	173,000.00 09-Oct-2015	139,600.00 12-Jun-2023	136,748.00 31-Dec-2023
Comments (achievements against targets):					



Household connections objective was met: 98 percent of end-target

Objective/Outcome: Community electricity connections constructed under the project

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Community electricity connections constructed under the project	Number	0.00 01-May-2015	1,800.00 09-Oct-2015	780.00 12-Jun-2023	319.00 31-Dec-2023
Community elec. connections constructed—Offgrid/minigrid—Only renewable sources	Number	0.00 01-May-2015	1,750.00 09-Oct-2015	780.00 12-Jun-2023	319.00 31-Dec-2023

Comments (achievements against targets):

Community Electricity connections were met only in 40,9 percent compared to the revised target. It is noted that the original target was much higher.

Objective/Outcome: People that obtain a daily and constant electricity service (through minigrids)

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
People that obtain a daily and constant electricity service (through minigrids) as a result of the project	Number	0.00 01-May-2015	42,000.00 09-Oct-2015	340.00 12-Jun-2023	339.00 31-Dec-2023



People that obtain a daily and constant electricity service and with prior access to the service, as a result of the project	Number	0.00 01-May-2015	33,000.00 09-Oct-2015	290.00 12-Jun-2023	288.00 31-Dec-2023
People that obtain a daily and constant electricity service (through minigrids) and without prior access to the service	Number	0.00 01-May-2015	9,000.00 09-Oct-2015	50.00 12-Jun-2023	51.00 31-Dec-2023
Comments (achievements against targets): Daily and constant service targets were met: 99,7 percent (99,3 and 102 percent for the breakdown indicators)					

A.2 Intermediate Results Indicators

Component: 1 - Renewable electricity service provision:(Cost \$215.61 M)

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Community electricity connections constructed under the project (schools)	Number	0.00 01-May-2015	1,200.00 09-Oct-2015	430.00 12-Jun-2023	263.00 31-Dec-2023
Comments (achievements against targets):					



Target for connections for schools was met in 61,2 percent compared to the revised figure. It is noted that the original target was much higher.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Community electricity connections constructed under the project (other public facilities)	Number	0.00	600.00	340.00	75.00
		01-May-2015	09-Oct-2015	12-Jun-2023	31-Dec-2023

Comments (achievements against targets):

Only 22 percent of the revised target was achieved. These results on community connections (this one and the previous) are consistent with the results obtained for the corresponding PDO indicators.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Generation Capacity of Renewable Energy (other than hydropower) constructed (MW)	Megawatt	0.00	10.00	5.00	5.00
		01-May-2015	09-Oct-2015	12-Jun-2023	31-Dec-2023
Generation Capacity of Renewable Energy constructed-Solar (MW)	Megawatt	0.00	8.00	5.00	5.00
		01-May-2015	09-Oct-2015	12-Jun-2023	31-Dec-2023



Comments (achievements against targets):

The two indicators were fully achieved, 5MW installed in both cases. It would appear that upon the absence of a third renewable energy source (other than solar or hydro) both indicators are measuring the same.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Micro-enterprises that obtained an electricity service under the project	Number	0.00 01-May-2015	4,220.00 09-Oct-2015	4,327.00 12-Jun-2023	6,870.00 31-Dec-2023
Individual micro-enterprises that obtained an electricity service under the project	Number	0.00 01-May-2015	4,200.00 09-Oct-2015	4,237.00 12-Jun-2023	6,870.00 31-Dec-2023
Micro-enterprises supported under the project that increase their production	Percentage	0.00	90.00		90.00

Comments (achievements against targets):

Project fully achieved by 158, 162 and 100 percent respectively. First two indicators appear have a significant overlap.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
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People that are satisfied with the benefits arising from obtaining access or an enhanced access to electricity through household connections	Percentage	0.00 01-May-2015	80.00 09-Oct-2015	80.00 12-Jun-2023	88.00 31-Dec-2023
Comments (achievements against targets): Target on people's satisfaction was surpassed, i.e. 110 percent achievement.					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
People that are satisfied with the benefits arising from obtaining access or an enhanced access to electricity through community connections	Percentage	0.00 01-May-2015	80.00 09-Oct-2015	80.00 12-Jun-2023	93.00 31-Dec-2023
Comments (achievements against targets): Target surpassed, 116 percent achievement of end-target					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion



People that obtained access to an electricity service for productive uses through electric fence	Number	5,768.00 15-May-2020	0.00 09-Oct-2015	27,480.00 12-Jun-2023	27,480.00 31-Dec-2023
Comments (achievements against targets): Access to electricity through electric fence was fully (and exactly) met – 100 percent					
Component: 3 - Project deployment support:(Cost \$6.71 M)					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Capacity building, outreach, and training activities implemented under the project	Number	0.00 01-May-2015	70.00 09-Oct-2015	70.00 12-Jun-2023	108.00 31-Dec-2023
Comments (achievements against targets): Training activities target was significantly surpassed – 154 percent					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Clean energy studies and	Number	0.00	10.00	4.00	4.00



strategies supported under the project		01-May-2015	09-Oct-2015	12-Jun-2023	31-Dec-2023
Comments (achievements against targets): Target for clean energy studies was fully met – 100 percent, though original end-target was much higher.					



B. KEY OUTPUTS BY COMPONENT

Objective/Outcome 1: To provide and enhance access to modern energy services in selected rural areas of Argentina.

Outcome Indicators

1. Direct project beneficiaries
2. People provided with access to electricity by household connections
3. Community electricity connections constructed under the project
4. People that obtain daily and constant electricity service through mini-grids.

Intermediate Results Indicators

1. Community electricity connections constructed under the project (schools)
2. Community electricity connections constructed under the project (other public facilities)
3. Generation Capacity of Renewable Energy (other than hydropower) constructed
4. generation Capacity of Renewable Energy Constructed – Solar
5. Micro-enterprises that obtained an electricity service under the project
6. Individual micro-enterprises that obtained an electricity service under the project
7. Micro-enterprises supported under the project that increased their production
8. People that are satisfied with the benefits arising from obtaining access or an enhanced access to electricity through household connections
9. People that are satisfied with the benefits arising from obtaining access or an enhanced access to electricity through community connections
10. People that obtained access to an electricity service for productive uses through electric fence



	11. Capacity building, outreach and training activities implemented under the project 12. Clean energy studies and strategies under the project.
Key Outputs by Component (linked to the achievement of the Objective/Outcome 1)	Component 1: Renewable electricity service provision 1. Electricity service for lighting, communication and other basic needs for isolated individual households and public facilities 2. Electricity service for lighting, communication and other basic needs for rural and isolated populations clusters and/or public institutions through mini-grids 3. Electricity service for potable water pumping for isolated individual households, public facilities and isolated communities. 4. Electricity service to micro-enterprises for the development of individual or collective productive uses activities. 5. Small works and goods as in-kind compensation under the relevant resettlement plans. Component 2: Solar thermal energy service provision (<i>this component was cancelled</i>) 1. Solar thermal energy in dispersed public facilities 2. Solar thermal energy for public buildings in rural areas. Component 3: Project deployment support 1. Market studies to determine potential energy demand and technologies at both urban and rural levels 2. M&E activities, including deployment of new technologies to remotely measure electricity consumption and use as well as subprojects performance 3. Communication strategies and outreach activities 4. Support for the development of pilot studies for technologies that have not been fully deployed in Argentina Component 4: Project management 1. Strengthening the operational capacity of the PCU and participating provinces and their respective PEUs



- | | |
|--|--|
| | <ol style="list-style-type: none">2. Project audits3. Training4. Financing travel costs for PCU and PEUs staff5. Rental of vehicles for project supervision activities. |
|--|--|

**ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION****A. TASK TEAM MEMBERS**

Name	Role
Preparation	
Lucia Spinelli	Task Team Leader(s)
Alvaro Larrea	Procurement Specialist(s)
Luz Maria Meyer	Financial Management Specialist
Cesar Adrian Arreola Croda	Team Member
Francisco Javier Obrique Arqueros	Team Member
Maria Pia Cravero	Counsel
Lilian Pedersen	Social Specialist
Elena Segura Labadia	Counsel
Fabiola Altimari Montiel	Counsel
Santiago Scialabba	Social Specialist
James Victor Pannett	Team Member
Noreen Beg	Social Specialist
Fernando J. Brunstein	Social Specialist
Enrique O. Crousillat	Team Member
Supervision/ICR	
Lucia Spinelli	Task Team Leader(s)
Maria Elizabeth Grandio, Francisco Rodriguez, Antonella Celeste Perila	Procurement Specialist(s)
Pedro Miguel Pinto Carvalho De Figueiredo	Financial Management Specialist
Mariana Echegaray	Team Member
Claudio Alvaro Mentasti	Procurement Team



Nahir Mailen Lamadrid	Team Member
Sofia Carla Gervasi	Procurement Team
Maria Gracia Lanata Briones	Social Specialist
Pablo Francisco Herrera	Environmental Specialist
Maria Lucrecia Bertelli	Social Specialist
Wendy Claribel Guerra Navarro	Procurement Team
Maria Pia Cravero	Counsel
Andrea Maria Castro Astudillo	Team Member
Claudio Luis Daniele	Environmental Specialist
Carolina Marcela Crerar	Team Member
Lilian Pedersen	Social Specialist
Maria Emilia Sparks	Social Specialist
Santiago Scialabba	Social Specialist
Alvaro Larrea	Procurement Team
Paula Genis	Team Member

B. STAFF TIME AND COST

Stage of Project Cycle	Staff Time and Cost	
	No. of staff weeks	US\$ (including travel and consultant costs)
Preparation		
FY13	12.882	64,657.03
FY14	19.431	77,658.28
FY15	20.385	112,146.65
FY16	.151	4,605.57
Total	52.85	259,067.53
Supervision/ICR		
FY15	0	13,583.62



FY16	17.084	113,476.62
FY17	27.054	149,751.88
FY18	14.025	99,748.44
FY19	9.309	67,827.26
FY20	29.606	133,972.26
FY21	23.517	99,208.05
FY22	18.841	148,011.08
FY23	15.700	145,653.82
FY24	23.388	162,608.91
Total	178.52	1,133,841.94

**ANNEX 3. PROJECT COST BY COMPONENT**

Components	Amount at Approval (US\$M)	Actual at Project Closing (US\$M)	Percentage of Approval (%)
1. Renewable electricity service provision	183.40	141.35	71.9
2. Solar thermal energy service provision	7.90	0	0.0
3. Project deployment support	5.70	1.74	30.5
4. Project management	3.00	2.29	76.3
Total	200.00	145.38	72.7

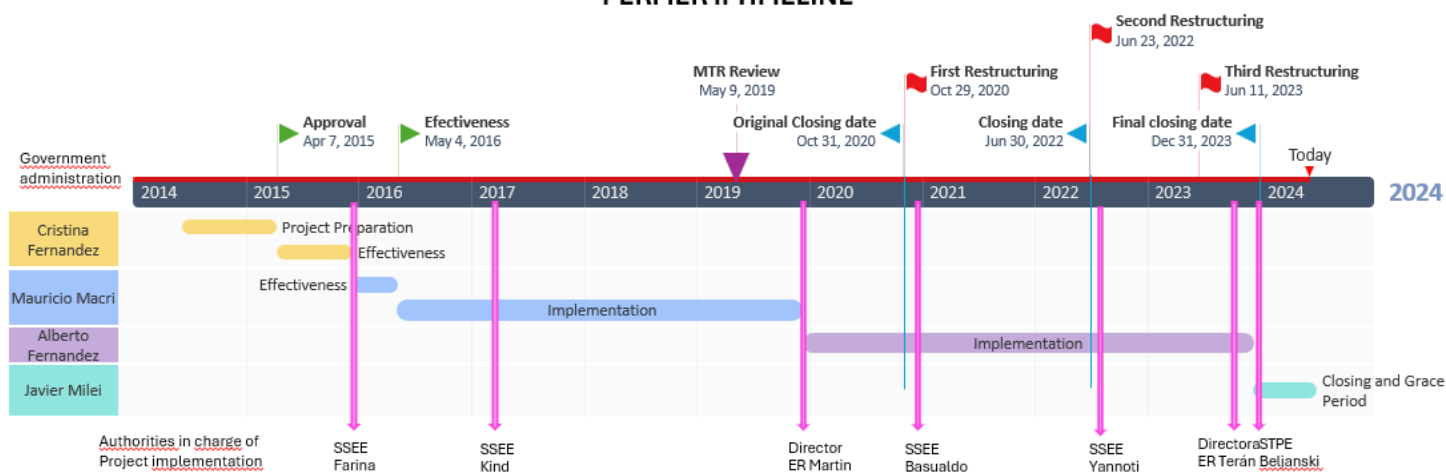
Figures correspond to IBRD financing. In addition, the provincial government of Salta contributed with US\$4.8 million in supporting Component 1, sub-project Luracatao, Municipality of Seclantas.

Actual figures for Component 1 and Total Costs will be reduced by an amount of around US\$4 million that the GoA will return to the Bank upon its decision to cancel twelve uncompleted contracts. The said amount corresponds to the advance payment made to contractors.



ANNEX 4. PROJECT TIMELINE

PERMER II TIMELINE



References

- Project phases
- ▼ Mid Term Review
- Restructurings
- ◀ Closing dates

29-Oct-2020	44.86	Change in Results Framework Change in Components and Cost Change in Loan Closing Date(s) Cancellation of Financing Reallocation between Disbursement Categories Change in Implementation Schedule Other Change(s)
23-Jun-2022	69.26	Change in Results Framework Change in Components and Cost Change in Loan Closing Date(s) Reallocation between Disbursement Categories Change in Implementation Schedule
11-Jun-2023	91.01	Change in Results Framework Change in Components and Cost Change in Loan Closing Date(s) Cancellation of Financing Reallocation between Disbursement Categories Change in Implementation Schedule



ANNEX 5. PROJECT OUTCOME – APPLICATION FOR THE SPLIT RATING

		Before Restructuring 2016-Oct. 2020	After Restructuring Oct. 2020-end 2023
Relevance of Objective		High	
Efficacy (PDO)		Negligible	Modest to Substantial
Efficiency		Modest to Substantial	
1	Outcome ratings	Unsatisfactory	Moderately Satisfactory
2	Numerical value of the outcome ratings*	2	4
3	Disbursement	\$44.86 million	\$46.15 million
4	Share of disbursement	0.493	0.507
5	Weighted value of the outcome rating (Row 2 x Row 4)	0.986	2.028
6	Final Outcome Rating	Moderately Unsatisfactory (0.986 + 2.028 = 3.014 rounded to 3)	

*Note: Highly Unsatisfactory (1); Unsatisfactory (2); Moderately Unsatisfactory (3); Moderately Satisfactory (4); Satisfactory (5); Highly Satisfactory (6)

Main considerations/assumptions:

- According to the Bank's ICR Guidelines, the split rating method is aimed at assessing the project's achievement across its lifetime, i.e. for different scenarios with and without the revision of objectives.
- The assessment is done for two periods defined by the scenarios prior to the first restructuring of 2020, and the remaining period until project closing. Methodological aspects, including the selection of the periods and targets assessed, are presented below.
- Efficacy ratings are explained as follows: **Negligible** for the first period as the PDO indicators were not achieved or barely achieved; and **Modest to Substantial** for the second period as three end targets were almost fully achieved but one performed poorly.
- High** and **Modest-to-Substantial** ratings are assigned for Relevance and Efficiency, respectively. That is, a single rating for each, irrespective of the period of analysis, due to the following reasons:
 - Relevance remained high throughout the entire implementation period.
 - The main efficiency variable, economic returns, is an indicator measurable at project completion, with no distinction between periods.
- Overall outcome ratings are justified as follows: **Unsatisfactory** for the first period as there were major shortcomings in the achievement of objectives. **Moderately Satisfactory** for the second period because there were moderate shortcomings in terms of efficacy and efficiency.



Methodological aspects:

- The split rating method takes into account original and revised objectives/outcome targets in deriving the project's overall outcome rating. Accordingly, one restructuring defines two periods (or scenarios) of analysis: with and without the revision of objectives. More than one restructuring would increase the number of periods to analyze.
- Bank guidelines indicate that the overall rating is based on a weighted assessment of each period's achievements, taking into account the share of disbursements of each period.
- The assessment of two periods –prior and after the first restructuring– is chosen, instead of four periods, because the latter option would imply neglecting completely the project's very positive achievements of the last months of implementation, since no disbursements were reported after the third restructuring. That is, a rigid application of the weighting approach would unduly punish the assessment of the project's achievements and introduce a major distortion in the overall rating⁴. To overcome this problem it was opted to merge the periods subject to restructurings.
- A two-period approach avoids the distortion mentioned above, thus yielding more realistic and fair results. The approach implies the following plausible assumption: that the period after the restructure of 2020 was a single period characterized by the progressive reduction of end targets (with some temporary fluctuations) and extensions of the closing date with view to adapt to changing conditions and enhance the project's chances to achieve its PDO.
- It is noted that the approach chosen does not alter the importance of the initial period of very poor performance (in terms of efficacy) since, whichever the case, it will keep the same significant weight of 49,3 percent⁵.
- Two approaches were applied in assessing the project's compliance of the original end-targets (efficacy): (i) against the actual achievement at the end of the first period (October 2020) and, alternately, (ii) against the achievement at closing date (December 2023)⁶. Both approaches yield poor results (see Table 3 in main text): negligible and modest-to-negligible, respectively. Hence, a conservative rating of **Negligible** is assigned.

⁴ Using the share of disbursements as a measure to assign weights among the periods being assessed can introduce a distortion in the analytical process, particularly for infrastructure projects –large or small– since, by nature, these operations are characterized by a lag between investments (disbursement) and outputs.

⁵ On the other hand, avoiding the distortion inherent to the weight system implies also avoiding a poorer rating for efficacy.

⁶ The first approach focusses on the project's achievement at the date that the original targets were meant to be achieved, i.e. assesses the project against its original targets and deadlines. The second approach incorporates the impact of the restructuring(s) –in this case the extension of the closing date– in facilitating the project's performance.



ANNEX 6. EFFICIENCY ANALYSIS

Economic Analysis

1. Following the appraisal's approach, the ex-post economic analysis focuses on the main infrastructure Component 1: Renewable electricity service provision, that encompassed, among other subcomponents, the acquisition and installation of stand-alone solar and wind systems and PV for isolated individual households and public facilities. In particular, the analysis focuses on the service to isolated households which accounted for 136,728 beneficiaries, i.e. 74.4 percent of the project's direct beneficiaries.
2. In order to compare ex-post results with the economic evaluation undertaken during project preparation, the analysis aims to replicate the PAD's approach. To this end, it considers the same assumptions for rural households' avoided costs, i.e. the substitution of expenses associated with traditional energy sources, such as kerosene lamps, candles, gas and batteries, as well as for estimates of the consumers' willingness to pay (WTP) for additional energy. Values for the economic benefits are estimated for the three SHS technological options used in the project: 2GCA and 2GCC and 3G, the latter a third-generation option that was not considered in the PAD's economic analysis.
3. 3G equipment is a low cost, basic service option that was made available to 90,916 beneficiaries. It offered the advantage of reaching a large number of beneficiaries within a relatively short time. However, its benefits are technically constrained, as well as its economic life. Accordingly, it is estimated that in comparison to the well-established 2G SHSs, 3G benefits are limited to the avoidance of existing costs and would not provide additional benefits associated to the beneficiaries' WTP for additional energy.
4. The economic internal rate of return (EIRR) of the electricity services provided to isolated individual households is 21,3 percent and its net present value (NPV) US\$18.2 million, based on a 10 percent discount rate that, as at appraisal, is considered to be the economic opportunity cost of capital. These positive results compare to an EIRR of 28.8 percent estimated at appraisal for the same component, and a NPV of US\$59.6 million. The lower economic indicators achieved by the project are attributed to both higher capital costs per unit of equipment and the considerable reduction in the size of the investment component of the project.
5. Considering that there is uncertainty around the 3G equipment's economic life, sensitivity analysis was undertaken for a shorter lifetime -from 10 to 6 years. It was found that in such case the EIRR would go down to 17.9 percent. Also, if the 3G is excluded, the project's EIRR would decrease to 13,7% thus confirming that the large incorporation of 3G equipment had a positive impact in the project's economic performance.
6. The table below presents the flow of costs and benefits for the project's isolated households solar component.



Table 6.1. Flow of costs and benefits for the project's isolated households solar component

	Capital Costs			Batt. Replace.		Benefits 1/			TOTAL COSTS	TOTAL BENEFITS	NET BENEFITS
	2G CA	2G CC	3G	2G CA	2G CC	2G CA	2G CC	3G			
u.cost	3320	1775	523			avoided cost	210	210	195		
#hh	166	11287	22729			WTP 2/	208	109	0		
	Capital Costs					Benefits					
0	0.55	20.03	11.89						32.47		-32.47
1						0.07	3.59	4.44	0.00	8.10	8.10
2						0.07	3.63	4.44	0.00	8.14	8.14
3						0.07	3.67	4.44	0.00	8.17	8.17
4						0.07	3.70	4.44	0.00	8.21	8.21
5				0.11	4.01	0.07	3.74	4.44	4.12	8.25	4.13
6						0.07	3.78	4.44	0.00	8.29	8.29
7						0.07	3.82	4.44	0.00	8.33	8.33
8						0.07	3.85	4.44	0.00	8.36	8.36
9						0.08	3.89	4.44	0.00	8.40	8.40
10				0.11	4.01	0.08	3.93	4.44	4.12	8.44	4.33
11						0.08	3.97		0.00	4.05	4.05
12						0.08	4.01		0.00	4.09	4.09
13						0.08	4.05		0.00	4.13	4.13
14						0.08	4.09		0.00	4.17	4.17
15				0.11	4.01	0.08	4.13		4.12	4.21	0.09
16						0.08	4.17		0.00	4.25	4.25
17						0.08	4.21		0.00	4.30	4.30
18						0.08	4.26		0.00	4.34	4.34
19						0.08	4.30		0.00	4.38	4.38
20						0.08	4.34		0.00	4.43	4.43
										NPV:	18.2
										IERR:	21.3%
1/	Benefit in US\$/year/household										
2/	Additional benefits associated to more and a better quality of energy source										



ANNEX 7. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

1. The PCU has prepared a thorough Project Final Report (May 2024) that addresses the project's background, development objectives, implementation issues, performance and sustainability aspects, as well as its compliance with environmental and social safeguards, financial management and procurement.

2. The report comprises three chapters: (1) context during project preparation, development objectives, components, conditions of IBRD loan agreement, results indicators and implementation arrangements; (2) implementation processes, paying special attention to Component 1: Renewable electricity service provision; and (3) results, implementation challenges and lessons learned.

3. The report's quantitative conclusions fully coincide with the ICR's, including the impact of the project restructurings on loan cancellations and the results framework, the level of execution and final costs, as well as the project's compliance with PDO and intermediary indicators. That is, the Final Report presents the same level of achievement as the ICR's.

4. The borrower's report elaborates on multiple challenges that compromised the normal implementation of the project and determined an uneven progress in time, including periods slow or negligible progress contrasting with periods of great advances. It presents three categories of implementation challenges: (1) administrative, financial and management; (2) impact of the COVID-19 pandemic; and (3) macroeconomic challenges. Most salient challenges were:

Administrative, financial and management

- Impact of changes in government authorities in late 2015, 2017 and 2019.
- Impact of several locations of the project within the GoA's administration (late 2015, 2017 and 2019).
- Procurement delays caused by the establishment of new administrative procedures (2016).
- A shift in SHS technology (2017) that required reaching consensus among stakeholders and revising the project's budget and results framework.
- Delays in appointing the PCU's General Coordinator (2019-2020).

COVID-19 pandemic

- Emergency measures given by the Federal Government as a health protection policy slowed down most project activities during 2020 and 2021.

Macroeconomic

- Price increases in minerals (copper, aluminum, iron), oil, polysilicon, glass and overall logistic activities.
- Impact of severe import restrictions on the normal execution of subprojects.

5. The report presents also a detailed and extensive set of lessons associated to the project's experience in dealing with management, technical, M&E and social and environmental issues. These include, among others, the need for a platform for the integrated management of contracts, improved certification of works, flexibility in agreements and operational manuals, acknowledging qualitative differences in the measure of some indicators (e.g. total number of beneficiaries), improved communications between the parties involved, a better dissemination of the GRM and capacity building in several areas, particularly on social and environmental matters for the PEUs. These lessons were taken into account in the formulation of Section V of this ICR.



ANNEX 8. SUPPORTING DOCUMENTS

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